Control Systems I — Achievement Test (Answer Sheet)

Name: ______________________ Student ID: ______________________

⚠ Instructions:

- No cheat sheet, calculator, or electronic device is allowed.
- You may use both sides of this sheet.
- Write your name and ID on every sheet.
- Clearly mark question numbers (e.g., Q1, Q2(2)).
- Show all intermediate steps; partial credit is given.
- Write all letters, numbers, and symbols clearly below — illegible writing may lose credit..

Q1. Since the damped torque $c\dot{\theta}$ is applied in the same direction of $\ddot{\theta}$, the resultant equation of motion is

$$ml^2\ddot{\theta} + c\dot{\theta} - mgl\sin\theta = \tau.$$

15pts if correct
5pts if wrong sign on $c\dot{\theta}$

Q2. (1) Around $\theta = 0$ and $\dot{\theta} = 0$, $\sin\theta \simeq \theta$ and $\tau = 0$, so $ml^2\ddot{\theta} + c\dot{\theta} - mgl\theta = 0$. Having $\boldsymbol{x} = [\theta, \dot{\theta}]^{\mathrm{T}}$,

$$\frac{d\boldsymbol{x}}{dt} = \begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{c}{ml^2}x_2 + \frac{g}{l}x_1\end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{c}{ml^2}\end{bmatrix}\boldsymbol{x},$$

5pts for correct matrix A
5pts for not oscillatory
5pts for instability

so $A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{c}{ml^2}\end{bmatrix}$ and the eigenvalues are $\frac{-c\pm\sqrt{c^2+4gl^3m^2}}{2l^2m}$. Since $c^2 + 4gl^3m^2 > 0$, they are real numbers, and motion is never oscillatory. One of the eigenvalues $\frac{-c+\sqrt{c^2+4gl^3m^2}}{2l^2m}$ is positive, so the system is unstable.

(2) Around $\tilde{\theta} = 0$ and $\dot{\tilde{\theta}} = 0$, $\sin\theta = -\sin\tilde{\theta} \simeq -\tilde{\theta}$, so $ml^2\ddot{\theta} + c\dot{\theta} + mgl\theta = 0$.

$$\frac{d\boldsymbol{x}}{dt} = \begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{c}{ml^2}x_2 - \frac{g}{l}x_1\end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{c}{ml^2}\end{bmatrix}\boldsymbol{x},$$

5pts for correct matrix A
5pts for oscillatory condition
5pts for stability

so $A = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{c}{ml^2}\end{bmatrix}$ and the eigenvalues are $\frac{-c\pm\sqrt{c^2-4gl^3m^2}}{2l^2m}$. When $c^2 - 4gl^3m^2 < 0$, the motion is oscillatory, and the real part is $-\frac{c}{2l^2m} < 0$. Otherwise, $-c \pm \sqrt{c^2 - 4gl^3m^2}$ are real numbers and always negative. So, the system is stable regardless of the parameter values.

(3) Around $\theta = \pi/2$ and $\dot{\theta} = 0$, $\sin\theta = \sin(\theta^* + \delta) \simeq \sin\theta^* + \cos\theta^*\,\delta \simeq 1$. With $\tau = 0$, EOM becomes $ml^2\ddot{\theta} + c\dot{\theta} + mgl = 0$, so this is not an equilibrium. Having $\tau = mgl$, equation of motion becomes $ml^2\ddot{\theta} + c\dot{\theta} + mgl = mgl \rightarrow ml^2\ddot{\theta} + c\dot{\theta} = 0$. Then $\boldsymbol{x} = [\theta^* + \delta, \dot{\delta}]^{\mathrm{T}}$,

5pts for not an equilibrium
5pts for $\tau = mgl$
5pts for correct matrix A
5pts for not oscillatory
5pts for instability

$$\frac{d\boldsymbol{x}}{dt} = \begin{bmatrix}\dot{\delta}\\ \ddot{\delta}\end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{c}{ml^2}x_2\end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{c}{ml^2}\end{bmatrix}\boldsymbol{x},$$

so $A = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{c}{ml^2}\end{bmatrix}$ and the eigenvalues are $0, \frac{-c}{2ml^2}$. These are both real numbers, so the motion is not oscillatory. The maximum value is 0, so the system is unstable.

Q3. a) The solution is given by $e^{At}\boldsymbol{x}_0$

5pts for indicating e^{At}, 15pts for correct answer

b) $e^{At}\boldsymbol{x}_0 = \begin{bmatrix} e^{\lambda_1 t} & 0 & 0 \\ 0 & e^{\lambda_2 t} & 0 \\ 0 & 0 & e^{\lambda_3 t}\end{bmatrix}\begin{bmatrix}1\\-1\\2\end{bmatrix} = \begin{bmatrix} e^{\lambda_1 t} \\ -e^{\lambda_2 t} \\ 2e^{\lambda_3 t}\end{bmatrix}$

5pts for indicating $e^{\lambda_i t}$, 15pts for correct answer

You may request additional sheet if necessary. Be sure your name and student ID appear on every page.